TEXORA · LECTURE SHEET

# Textile Recycling

*Returning textile waste to production — objectives and benefits, the manufacturing processes of mechanical and chemical recycling from collection to new yarn, and the impacts on industry and environment.*

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 8 · LEVEL 4 TERM 1

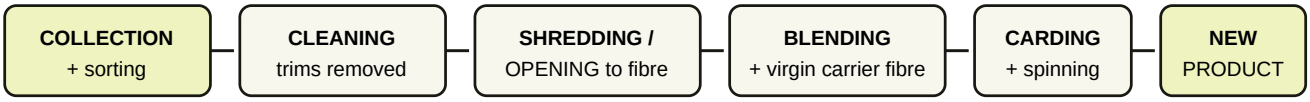
## 1. Where textile waste arises

Textile waste divides by origin. **Pre-consumer (post-industrial) waste** arises inside factories: spinning fly and noil, weaving selvedges and cut ends, dye-house off-cuts, garment cutting waste (often 10–20 % of fabric issued), rejected fabric and end-of-roll remnants. It is clean, of known composition, and already collected — the easiest recycling feedstock. **Post-consumer waste** is discarded clothing and household textiles: valuable in volume but mixed in fibre, soiled, and fitted with trims (zips, buttons, prints) that recycling must remove. A third stream, **process chemicals and water**, is treated under environmental engineering (TE-4204) rather than here.

## 2. Objectives and benefits

- **Resource recovery:** return fibre — the most resource-expensive part of a textile — to production instead of landfill or burning.
- **Raw-material cost:** recovered fibre and reclaimed material are usually cheaper than virgin equivalents for suitable end-uses.
- **Waste-disposal saving and compliance:** mills avoid disposal cost and meet buyer and regulatory requirements on waste handling; many buyers now audit waste destinations.
- **Environmental load reduction:** every kilogram of fibre recycled is a kilogram not grown, polymerised, dyed from scratch or landfilled — with corresponding savings in land, water, energy and emissions, and avoided methane from landfilled natural fibres.
- **New products and markets:** shoddy and mungo (recovered wool fibre), recycled-cotton yarns, nonwoven insulation, felts, wiping cloths, and recycled-content clothing lines.

## 3. The manufacturing process of recycling




Mechanical recycling route — fibre length shortens at every pass, so end-uses step down in demand

Chemical route (for synthetics and cellulose): dissolve / depolymerise → purify → re-extrude or re-spin at near-virgin quality

Fig. 1 — The recycling chain: collection and sorting decide everything downstream; a badly sorted input cannot be repaired by good machines.

## 3.1 Mechanical recycling — step by step

1. **Collection and sorting** — by fibre type (cotton, wool, polyester, blends), by colour (sorting by colour avoids re-dyeing), and by condition. Sorting is the quality gate of the whole process.
2. **Cleaning and trim removal** — dust and soil removed; zips, buttons, labels and prints cut away.
3. **Cutting and shredding/opening** — fabric is cut into small pieces and torn apart on pinned opening machinery (the traditional wool trade's shoddy/mungo machines) back into fibrous form. Each tearing pass shortens fibres and lowers their strength — the physical reason recycled fibre is weaker than virgin.
4. **Blending** — recovered fibre is blended with a proportion of virgin "carrier" fibre (longer, stronger) to restore spinnability; typical yarn blends carry a minority-to-majority share of recycled content depending on the count and end-use.
5. **Carding and spinning** — usually on rotor (open-end) systems, which tolerate shorter fibres better than ring spinning, into coarse-to-medium counts for denim, towels, blankets, furnishing and workwear fabrics — or into nonwovens (felts, insulation, padding) where no spinning is needed at all.

## 3.2 Chemical recycling

For polyester, chemical recycling depolymerises the polymer (by glycolysis, methanolysis or hydrolysis) back to monomers or intermediates, purifies them, and re-polymerises — yielding fibre of essentially virgin quality, at higher cost and with chemical-plant scale. For cellulosics, dissolution routes (of the lyocell family) dissolve waste cotton into pulp from which new regenerated fibre is spun. Chemical routes break mechanical recycling's quality ceiling but demand clean, single-polymer feedstock — which returns the process, again, to sorting.

**WORKED EXAMPLE — SORTING YIELD OF COLLECTED TEXTILES**

A collection unit receives 1,000 kg of post-consumer textiles. Sorting gives: directly reusable garments 15 % → **150 kg**; cutting into wiping cloths 25 % → **250 kg**; fibre recovery (shredding)

45 % → **450 kg**; unrecoverable residue (soiled, mixed trims, rubberised items) 15 % → **150 kg**. Even before any spinning, 85 % of the input has a route — but note the value order: reuse preserves the whole garment's embedded work; wiping cloths preserve the fabric; fibre recovery preserves only the raw material. Recycling policy and plant design both follow this hierarchy: **reuse > remanufacture > fibre recovery > energy recovery > landfill**.

## 4. Impact on industry

For industry, recycling converts a disposal cost into a product line and a compliance asset; it creates demand for collection, sorting and opening machinery and for recycled-content certification; it exposes mills to feedstock variability (recycled input is inherently less uniform than virgin bales, so blending and process control — Topic 7 — must work harder); and it is increasingly a **market-access condition**, as buyers set recycled-content targets and regulators in export markets move toward extended producer responsibility and recycled-content labelling.

## 5. Impact on environment

Environmentally, textile recycling avoids landfill volume and the methane generated by buried natural fibres, displaces virgin fibre production with its land, irrigation, fertiliser and polymerisation burdens, and (for colour-sorted mechanical recycling) avoids whole dyeing cycles. Its honest limits: collection and transport consume energy; shredding degrades fibre so products are often *downcycled* (a shirt becomes insulation, not another shirt); blends and elastane contaminate feedstock; and chemical routes carry their own solvent and energy footprints. Recycling is a major improvement on disposal, not a licence for unlimited throughput — the hierarchy in the worked example keeps prevention and reuse above it.

## Glossary

**Pre-consumer waste** — factory waste arising before the product reaches a user. **Post-consumer waste** — textiles discarded after use. **Shoddy / mungo** — fibre recovered by shredding wool and wool-blend materials. **Mechanical recycling** — recovery by sorting, cleaning, shredding and re-spinning. **Chemical recycling** — recovery by depolymerisation or dissolution to near-virgin feedstock. **Downcycling** — recycling into a lower-demand application because quality fell in processing. **Carrier fibre** — virgin fibre blended with recovered fibre to restore processability. **Extended producer responsibility** — policy making producers responsible for products' end-of-life.

## Key points

• Sorting is the quality gate; pre-consumer waste is the easiest feedstock, post-consumer the largest.
• Mechanical route: collect → sort → clean → shred/open → blend with carrier fibre → card → spin (usually rotor) or nonwoven.
• Worked sorting: of 1,000 kg — 150 reuse, 250 wiping, 450 fibre recovery, 150 residue.
• Chemical recycling restores near-virgin quality for suitable single-polymer streams, at higher cost.
• Benefits are real but bounded: fibre shortens, products downcycle, blends resist separation.

## Practice

1. A recycling plant processes 2,500 kg/day of cutting waste with the sorting yields of the worked example's fibre-recovery mindset: 8 % residue, the rest to fibre recovery after removing 12 % (of input) as reusable remnants. Compute the daily fibre-recovery mass.
2. Why is colour sorting economically valuable in mechanical cotton recycling?
3. Explain why recycled cotton is normally blended with virgin or synthetic carrier fibre, in terms of what shredding does to fibres.
4. State the waste hierarchy and place these in it: (a) exporting used garments for resale, (b) shredding worn denim into insulation, (c) burning mixed textile waste for its heat in a boiler with no energy recovery credit claimed beyond heat, (d) landfilling.

### Answer key

1. Reusable remnants = 12 % × 2,500 = 300 kg. Residue = 8 % × 2,500 = 200 kg. Fibre recovery = 2,500 − 300 − 200 = 2,000 kg/day.
2. Colour-sorted recovered fibre can be spun and used in its existing colour, skipping the entire dyeing process for that material — saving dye, chemicals, water, energy and the shade-matching risk — and colour families command better prices than mixed "grey" output, which must be dyed dark or used invisibly. Sorting by colour therefore converts a cost (dyeing) into margin.
3. Shredding tears fabric apart mechanically, and every tearing pass breaks fibres shorter and damages their surfaces, lowering staple length, strength and spinning cohesion. Yarn spun from 100 % recovered fibre is therefore weak, uneven and limited to very coarse counts. Blending with longer, stronger virgin or synthetic carrier fibre restores the blend's average length and strength distribution so that commercially useful counts can be spun; the recycled share is set as high as the end-use's strength requirement allows.
4. Hierarchy (best to worst): reuse > remanufacture/recovery of material > fibre recovery > energy recovery > landfill. (a) Reuse — top. (b) Fibre recovery into a product (material recovery, albeit downcycled). (c) Energy recovery — above landfill, below material recovery. (d) Landfill — last.